

Laksh Gupta

📧 he/him ✉️ lakshgupta.phy@gmail.com 📍 Bonn, Germany 🇮🇳 Indian

Master of Science in Astrophysics student at University of Bonn, Germany, through the scholarship program of the [Bonn-Cologne Graduate School of Physics and Astronomy](#). I studied White Dwarfs using data from the Hubble Space Telescope for my undergraduate thesis supervised by [Professor Samyaday Choudhury](#) (Ahmedabad University, India) and in collaboration with [Dr. Annalisa Calamida](#) (Space Telescope Science Institute, USA).

EDUCATION

Master of Science in Astrophysics, [University of Bonn](#), Germany October 2025 - October 2027

Bachelor of Science (Honors) in Physics, [Ahmedabad University](#), India August 2020 - December 2024

- GPA - 3.32/4
- Minor in Mathematics.
- Relevant Coursework: Quantum Mechanics, Atomic & Nuclear Physics, Condensed Matter Physics, Plasma Physics, Nonlinear Dynamics, Computational Mathematics, Electronics, Quantum Computing, Complex Analysis, Mathematical Statistics, Linear Algebra, Differential Equations, Multivariate Calculus, Advanced Writing

12th Grade, [La Martiniere College](#), Lucknow, India March 2020
Score - 81.2%

10th Grade, [La Martiniere College](#), Lucknow, India March 2018
Score - 84.8%

RESEARCH EXPERIENCE

Research Intern, Astronomy & Astrophysics Group June 2025 - August 2025
Supervisor: [Professor Blesson Mathew](#) [Christ University, Bengaluru, India](#)

- Working in collaboration with [Arun Roy](#) (Indian Institute of Astrophysics) and [Shridharan Baskaran](#) (Tata Institute of Fundamental Research), I worked on modelling a Herbig Be star with boundary layer accretion mechanism in [RADMC3D](#) to understand forbidden [Ne II] and [Ne III].

Summer Research Fellow April 2025 - June 2025
Supervisor: [Professor Blesson Mathew](#) [Science Academies' Summer Research Fellowship](#)

- Conducted a detailed theoretical and observational review on accretion mechanisms in Herbig Ae/Be stars, identifying a switch from magnetospheric to boundary layer accretion.
- Prepared a Python pipeline to produce the largest and most complete catalogue of identified Herbig Ae/Be stars from literature.
- Initiated modeling efforts using RADMC3D to study switch in accretion mechanism in Herbig Ae/Be stars, generating SEDs of which match the inclination effects and silicate features in Herbig-like systems.

Research Intern, Astronomy & Astrophysics Group May 2024 - September 2025
Supervisor: [Professor Samyaday Choudhury](#) [Ahmedabad University](#)

- Working in collaboration with Dr. Annalisa Calamida to refine the star count-crossing time analysis technique developed during the undergraduate thesis.
- This work is under review at the [The Astrophysical Journal](#).

RESEARCH PROJECTS

Undergraduate Thesis: Study of White Dwarfs in the Globular Cluster NGC2808 July 2023 - May 2024

Supervisor: Professor Samyaday Choudhury Ahmedabad University

- Performed star count-crossing time analysis of different evolutionary phases using photometric data of NGC2808 from the HUGS survey.
- Overplotted various BaSTI models (White Dwarf Cooling Models, Isochrones and Evolutionary Tracks) on NUV-optical Color-Magnitude Diagrams to calculate crossing times across different evolutionary phases.
- Created a Python pipeline to systematically analyze stellar populations (White Dwarf Stars, Red Giant Branch Stars, Main Sequence Turnoff Stars) of NGC2808, potentially applicable to other globular clusters.
- *Our results indicate that the White Dwarf Sequence in NGC2808 has no bimodality up to 24 magnitude in F275W.*

Summer Intern May 2023 - October 2023

Supervisor: Professor Navinder Singh Physical Research Laboratory, Ahmedabad, India

- Gained hands-on knowledge of Nuclear Magnetic Resonance (NMR) apparatus.
- Built an NMR apparatus using minimum circuitry and instruments for academic teaching purposes.

Stellar Structure & Evolution April 2023 - July 2023

Supervisor: Professor Samyaday Choudhury Ahmedabad University

- Acquired foundations in stellar structure and evolution - particularly in low-mass stars, radiative transfer and nucleosynthesis in stellar interiors, globular clusters and open clusters, compact objects like white dwarfs, stellar collisions, and binary stars.
- Read *An Introduction to Modern Astrophysics* by Carroll and Ostlie and *Astrophysics for Physicists* by Arnab Rai Choudhuri.

SKILLS

Astronomy	BaSTI and MIST Models (White Dwarf Cooling Models, Isochrones, Evolutionary Tracks), Optical and Near-IR Hubble Data Analysis, Color-Magnitude Diagram Analysis, Reddening Estimation
Tools	RADMC3D, Vizier, xMatch, TOPCAT, TeX, Tracker
Programming Languages	Python (NumPy, SciPy, Pandas, Astropy, Shapely, Seaborn), MATLAB, Java, R
Communication	Fluent in English (IELTS 7 band) and Hindi

ATTENDED SCHOOLS & CONFERENCES

43rd Meeting of the Astronomical Society of India 15th - 19th February 2025

Poster Presentation National Institute of Technology, Rourkela, India

National School on Space Radiation 16th - 20th September 2024

Manipal Center Natural Sciences, MAHE, Manipal, India

School on Solar Physics 2nd - 6th September 2024

Physical Research Laboratory, Ahmedabad, India

OTHER PROJECTS

Profiling a Helium-Neon LASER beam December 2022

Optics Laboratory Project

- Designed the experimental apparatus to profile the He-Ne LASER beam profiling, ensuring precise alignment and calibration of optical components.
- Conducted a detailed error analysis of the experiment.

Need Analysis for Emotional Health and Well-being of students

May 2022 - September 2023

Supervisor: [Professor Shilpa Pandit](#)

Research Project

- Designed a comprehensive survey to collect data on various aspects of students' mental health ensuring inclusively.
- Cleaned data to remove inconsistencies and converted qualitative responses into numerical values using Python.
- The publication leading to this work is [Pandit et al. 2025](#).

Intrinsic Magnetic Field inside Neodymium Magnets

December 2021

Electromagnetism Laboratory Project

- Calculated the angular velocity (ω) of the neodymium magnets from the nail-motor experiment apparatus using Tracker software.
- Using ω measurements, the intrinsic magnetic field of the magnets was determined, and a thorough error analysis was conducted.

Newton's Cradle

December 2020

Classical Mechanics Laboratory Project

- Calculated the translational velocities of the bobs from the Newton's Cradle setup using Tracker software.
- After performing error analysis, translational velocity measurements were utilized to understand conservation of energy and momentum.

AWARDS & ACHIEVEMENTS

- Awarded the [Bonn-Cologne Graduate School of Physics and Astronomy Scholarship](#) 2025.
- Awarded the [Indian Science Academies' Summer Research Fellowship](#) 2025.
- **Qualified** [IIT-Joint Admission Test for Masters \(IITJAM\)](#) 2025 with an **All India Rank 981** (out of 16,000).

LEADERSHIP AND VOLUNTEERING ACTIVITIES

- *Peer Tutor* for PHY315 - Atomic & Nuclear Physics (January 2024 - May 2024) and COM102 - Advanced Writing (August 2021 - December 2021)
- Organized a [Math Fest](#) at Ahmedabad University (November 2023)
- Founded *Ramanujan Math Club* promoting student problem-solving and mathematics at Ahmedabad University (August 2023 - May 2024)
- Represented Physics at Ahmedabad University's [Career Development Centre](#) (May 2023 - May 2024)
- *Student Mitr* - Mentored 15 incoming students to acquaint them with Ahmedabad University's culture (August 2021 - April 2022)
- *Junior Manager* Outgoing Social Sector - Association Internationale des Étudiants en Sciences Économiques et Commerciales ([AIESEC](#)) in Ahmedabad (August 2021 - October 2021)
- Volunteered at [Prabhat Foundation](#) during the COVID-19 pandemic (May 2021 - July 2021)